Paleomagnetism of the Basement Rocks, Wichita Mountains, Oklahoma

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Measurement of paleomagnetism in the basement rocks of the Wichita Mountains, Oklahoma (Cambrian age), resulted in a virtual geomagnetic pole (VGP) position of 33.5°N, 135°E, which correlates with VGP's of Carboniferous rocks in North America. The remanent magnetization (RM) of greatest reliability is a CRM due primarily to secondary hematite derived from magnetite and/or titanomagnetite. Mineralogical studies indicate that the time of CRM acquisition can be correlated with a low-temperature metamorphism accompanying the uplift of the Wichita Mountains. The paleomagnetic result agrees with geological observations that the Wichita Mountains were uplifted in Pennsylvanian time.

INTRODUCTION

Recent radioisotope age determinations give an age of the basement rocks of the Wichita Mountains, Oklahoma, as 520 to 560 m.y. [*Tilton et al.*, 1962]. Earlier, the rocks had been mapped as late Precambrian by *Chase et al.* [1956] and of about the same age as the basement rocks of the St. Francois Mountains, Missouri [*Tilton et al.* 1962; *Hays and Scharon*, 1966]. The purpose of the paleomagnetic investigation was to determine the VGP position of the ancient magnetic field that magnetized these basement rocks in the light of their radioisotope age determinations, and it was therefore necessary to investigate the stability and reliability of the paleomagnetic results in terms of their magnetic properties and mineral composition.

GEOLOGY AND SAMPLING

Figure 1 shows the major rocks of the basement complex of the Wichita Mountains. These rocks, covering an area of approximately 1000 km², have been divided into three groups, the Carlton rhyolite group, the Wichita granite group, and the Raggedy gabbro group, by *Ham et al.* [1965]. The gabbro is considered to be the oldest, and the granite is found to be the intrusive equivalent of the extrusive rhyolite. This basement complex consists of layered, stratiform flows and sills which have been block-faulted.

A total of 48 samples were collected from 19 sites (Figure 1). Where samples had been taken for radioisotope dating, more than 5 oriented rock samples from each site were collected. An exception is site 15, which is about 3 km south of the age-dating location but is from the same intrusive body. Samples from site 19, being too weak magnetically for the spinner magnetometer used ($<10^{-6}$ cgs emu), were eliminated, and only 43 samples from 18 sites were used for paleomagnetic studies.

Each sample was oriented in situ with respect to geographic north and the horizontal plane. To eliminate the effects of local magnetic polarization (causing errors of as much as a 20° to 30° deflection from magnetic north) the orienting compass was raised about 1½ meters above the ground surface to check the north direction (an error of less than 3° was to be expected). Three to four cylindrical specimens, each 2.5 cm long and 2.5 cm in diameter, were drilled and cut from each sample.

PALEOMAGNETIC RESULTS

The RM for each specimen was measured with a 10-cps spinner-type magnetometer [*Johnson*, 1938]. The specimens were then partially demagnetized in an alternating field of 200 oe [*Doell and Cox*, 1963] and the RM remeasured. The value of 200 oe was chosen after a series of test demagnetizations were made in fields of 25, 50, 100, 200, 400, 600, or 800 oe with 10 representative specimens from different sites. These test demagnetizations showed that the soft components of the NRM could be removed by an alternating field of 200 oe. The Koenigsberger ratio Q was calcu-

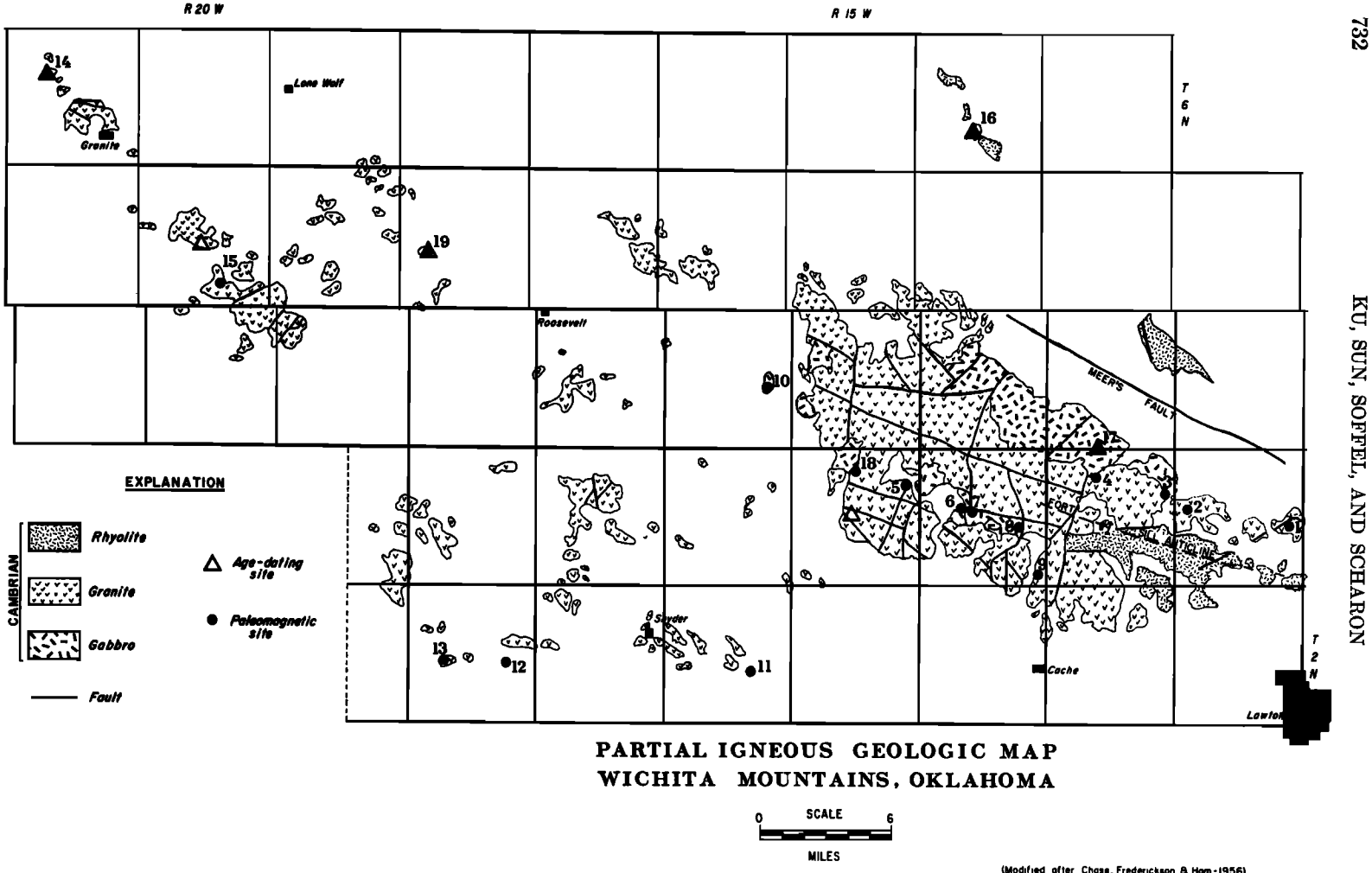


Fig. 1. Partial igneous geologic map, Wichita Mountains, Oklahoma, showing locations of paleomagnetic sample sites and age-dating sites of *Tilton et al.* [1962].

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TABLE 1. Paleomagnetic-Geologic Data and Mineral Ages for the Basement Rocks of the Wichita Mountains, Oklahoma (34.8°N to 99.9°W)

Site No.	M_n N	k_s ($\times 10^{-3}$ cgs)	M_n Q	M_n D	M_n I	M_n R	M_n α_{95}	M_{af} N	M_{af} Q	M_{af} D	M_{af} I	M_{af} R	M_{af} α_{95}	VGP Latitude °N	VGP Longitude °E	Iron Oxides, % M	H	I	U	Rock Types	Curie Temperatures between 20°C to 600°C Major	Minor	Age, m.y.* Rb/Sr
1†	8(2)	0.892	0.2	182.5	−39.6	2.158	45.2	8	0.4	150.3	−7.9	7.616	11.6	48.7	129.5	0.5	1.2	t		Gr	582	{236–259 455–478}	
2†	4(1)	0.487	0.2	113.5	−3.1	3.562	26.8	4	0.2	138.6	7.2	3.855	15.4	35.4	135.0	0.6	0.7			Gr			
3	4(1)	0.104	0.5	274.0	75.2	3.993	3.4	4	0.1	348.8	60.7	3.884	13.8			t	1.3			Gr			
4	4(1)	2.245	2.0	288.5	20.4	3.945	9.5	4	0.5	278.9	5.6	3.901	12.7			t	t	1.2		Gr			
5†	4(1)	0.132	10.3	113.2	−21.9	3.999	1.6	4	0.4	141.9	−4.0	3.997	2.3	41.7	136.7	t	0.8			Gr			
6	3(1)	0.416	169.4	197.4	−41.2	2.949	12.8	3	112.1	211.9	−38.8	2.936	14.5			0.5	t	0.5	t	Gb			
7	4(1)	0.400	226.5	228.8	−24.3	3.959	8.2	4	144.0	229.1	−24.8	3.962	7.9			t		0.1		Gb			
8	3(1)	0.235	10.8	313.1	18.3	2.982	7.6	3	0.3	346.6	2.2	2.921	16.0			t	t	0.2		Gb			
9	4(1)	0.011	15.1	330.9	−10.6	3.949	9.2	4	5.7	337.8	27.5	3.893	13.2			t	0.1	t		Gr			
10	8(2)	0.682	36.9	11.3	52.8	5.215	31.2	8	7.8	130.3	12 5	0.880	49.9			t		0.6		Gb	569		
11	4(1)	2.305	3.4	287.4	−18.5	3.997	2.4	4	1.4	275.8	−6.9	3.999	1.1			1.0		t		Gr	578	264,551	
12	4(1)	0.093	23.8	3.3	−56.1	3.987	4.7	4	10.2	7.9	−22.4	3.989	4.2			t	1.0	0.1		Gr			
13†	8(2)	0.027	0.5	108.2	46.7	7.420	14.3	8	0.3	105.8	26.3	7.353	15.1	4.6	150.6	t	0.3	t		Gr			
14	20(5)	0.670	1.1	313.5	48.3	17.081	12.3	20	0.2	310.0	26.4	10.927	21.6			0.6	t	0.2		Gr	575	136–344	520 ± 30
15†	20(5)	0.029	0.7	130.5	53.1	19.437	5 4	20	0.3	132.9	28.1	18.692	8.2	23.1	131.3	t	1.0			Gr			500
16†	24(6)	0.527	39.4	30.5	76.5	13.810	19.0	16(4)	0.8	143.4	48.7	15.198	8.1	16.9	113.8	t	2.5			Ry	583	165–578	
17†	36(9)	6.415	12.8	96.4	36.8	2.467	22.8	8(2)	0.2	143.8	−4.0	7.550	12.6	43.0	134.8	2.5	t	2.5		Gb	572	561	535 ± 30
18†	8(2)	0.561	0.5	66.5	68.7	7.924	5.2	4(1)	0.1	144.3	−3.0	3.774	19.2	43.0	133.9	0.4	0.6	t		Gr	573	265	
19	20(5)	Too weak to be measured																		Gr			520
Mean of the sites										138.0	11.4	7.381	14.7	33.4	133.9								

* After *Tilton et al.* [1962].
† Site mean used to calculate mean of the sites.
N, number of specimens measured. Number in parentheses indicates number of samples.
k_s, magnetic susceptibility.
Q, Koenigsberger ratio = $J(H)/(0.55 \times k_s)$; $J(H)$ is the intensity of remanent magnetization, 0.55 is the intensity of earth's total field for the present area.
D, declination of remanent magnetization (RM).
I, inclination of RM.
R, magnitude of the resultant of the *N* unit vectors.
α_{95}, semiangle of cone of 95% confidence (Fisher) for mean direction.
M_n, NRM.
M_{af}, RM after partial demagnetization in alternating field at 200 oe.
VGP, virtual geomagnetic pole for north magnetic pole ('reverse polarity').
M, magnetite and/or titanomagnetite.
H, hematite.
I, ilmenite.
U, ulvöspinel.
t, trace amount (<0.1%).
Gr, granite.
Gb, gabbro.
Ry, rhyolite.

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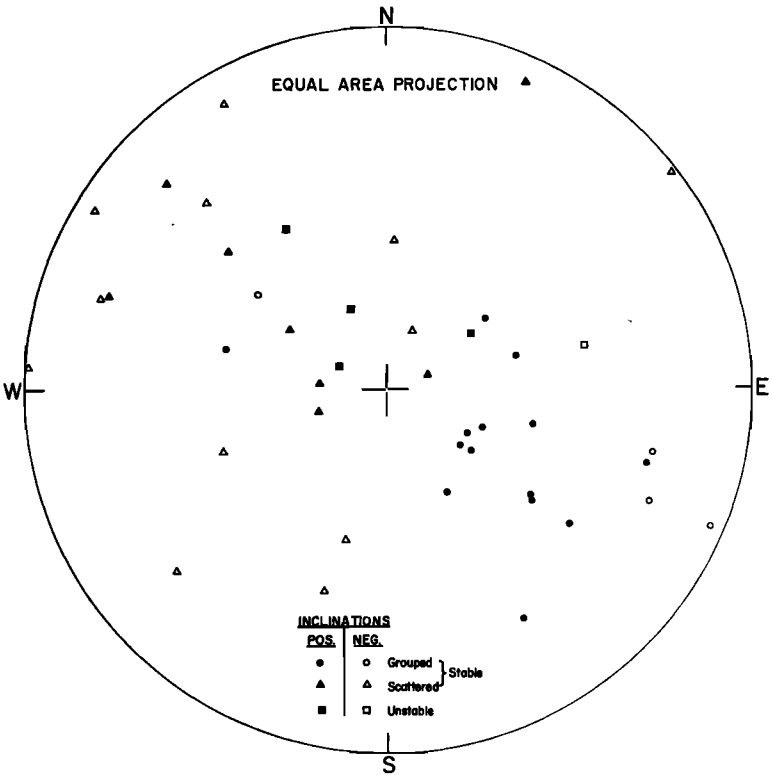


Fig. 2. Mean RM directions of samples before demagnetization.

lated before and after the ac 200-oe demagnetization. The induced magnetization, $J_i = \chi H$, of the samples was determined by multiplying the suspectibility χ of the samples (measured with a commercial susceptibility bridge) by the field strength of 0.55 oe as inferred for the sampling location from the 'Total Intensity of the Earth's Magnetic Field' map for 1965 (Hydrographic Office, Washington, D. C.). Table 1 shows the results of the magnetic measurements combined with the microscopic results and the age dating [*Tilton et al.*, 1962].

Figures 2 and 3 show the mean RM directions of the different samples before and after demagnetization. To test the reliability of the paleomagnetic results, we calculated the mean RM direction and α_{95} [*Fischer*, 1953] after ac demagnetization with 200 oe for each sample. The results from 5 samples (squares in Figure 3), each with 4 specimens, were discarded because their α_{95} exceeded 25°, which indicated that the samples were paleomagnetically unstable. The remainder of the stable samples showed that some are well grouped together (circles in Figure 3), but more than half of the paleomagnetically stable samples are scattered (triangles in Figure 3) in their RM directions after ac demagnetization. The lack of recognizable features for the block faults at the sites where the 'scattered' samples were taken makes it difficult if not impossible to assess the influence of the faulting. Therefore these samples (triangles in Figure 3) were not used in computing the VGP.

Additional magnetic and microscopic properties of those samples indicating stability and/or good grouping were examined to ensure the reliability of their final choice. The Q ratios for some of the sites (Table 1) reveal a rapid decrease after ac demagnetization. Some of these samples show good grouping of their RM directions after demagnetization (circles in Figure 3). This rapid decrease of the Q ratio indicates the presence of some magnetically unstable components of the RM, probably an isothermal and/or other viscous RM, that could

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be removed by the applied demagnetization alternating field of 200 oe [*Cox and Doell*, 1960]. The scatter of the rest of the samples increased rather than decreased by ac demagnetization (squares and triangles in Figure 3).

Microscopic studies (Table 1) show that all the samples contain more or less of a magnetic spinel phase mineral (presumably magnetite and/or titanomagnetite, but definitely not maghemite). However, the well-grouped samples generally contain a dominant amount of hematite in contrast to the scattered or unstable samples which tend to contain ilmenite as the dominant iron oxide. The acidic igneous rocks (granite and rhyolite) yield better paleomagnetic results than the basic igneous rocks (gabbro). The former contain, for the most part, hematite, and the latter contain ilmenite as the dominant iron oxide. The hematite content of the grouped samples is mainly secondary and is derived from magnetite and/or titanomagnetite. Almost all the hematite was produced by oxidation (martitization), yet the occurrence of secondary hematite due to unmixing cannot be excluded because a few of the hematite lamellas show typical exsolution characteristics. The silicate minerals show no indication of metamorphism, which suggests that the martite formed at low temperatures.

Curie points were measured for representative samples from the age-dated sites by determining the dependency of initial permeability of the samples on temperature between 20 and 600°C with a high-frequency Curie point bridge as described by *Fraunberger* [1955] and as adapted by *Petersen* [1965] for ferrimagnetic minerals. Plots of the measured data involving Curie point determinations for gabbro, granite, rhyolite, and synthetic magnetite are shown in Figure 4. The gabbro shows the presence of a major magnetic phase with a Curie temperature of 572°C, for which a unit-cell dimension of $a = 8.402$ A was determined by the X-ray powder method. This phase is

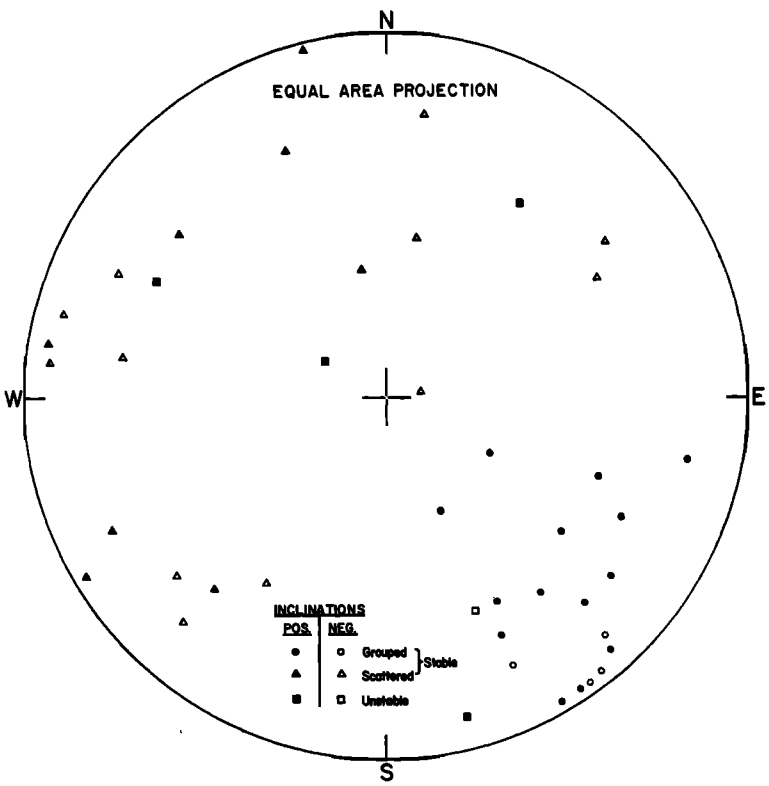


Fig. 3. Mean RM directions of samples after ac demagnetization with 200 oe.

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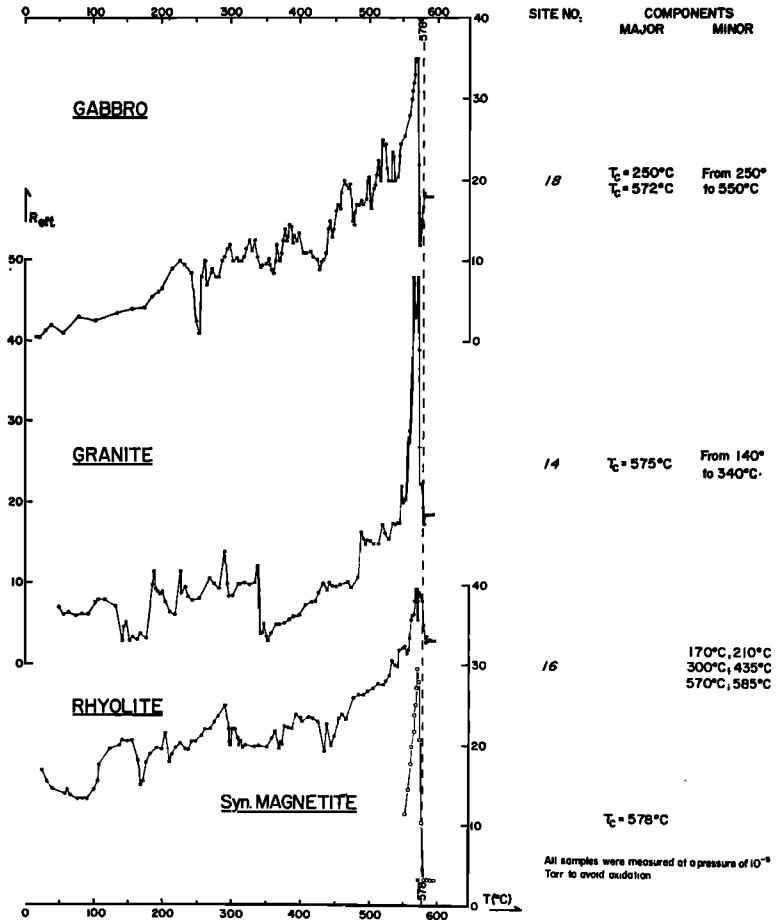


Fig. 4. Curie temperature measurements with a high-frequency Curie point meter. The effective resistance R_{eff} of the sample coil is plotted versus temperature.

assumed to be almost pure magnetite. The plot also indicates that several minor components have Curie temperatures between 250 and 550°C, the phase having a Curie temperature of 250°C being the most dominant one. The major magnetic phase for the granite sample has a Curie temperature of 575°C. A unit cell dimension of $a = 8.403$ A was measured for this component, and it is probably almost pure magnetite. The plot reveals that several minor components have Curie temperatures between 140 and 340°C. The rhyolite sample, that contained almost entirely hematite as iron oxide, had only small amounts of magnetic minerals with Curie temperatures of 170, 210, 300, 435, 570, and 585°C. For comparison, a measurement for synthetic magnetite is plotted with a Curie temperature of 578°C.

The Curie point measurements and polished section studies indicate that the NRM of the rock samples is carried partly by a cubic phase, which is probably almost pure magnetite, and partly by secondary hematite and by several minor magnetic phases which have Curie temperatures between 140°C and 550°C, having smaller saturation magnetization values than pure magnetite.

From the polished section studies combined with the ac-demagnetization spectrums and the Curie point measurements it is concluded that the RM of the grouped samples is a CRM (mainly that of the secondary hematite) that

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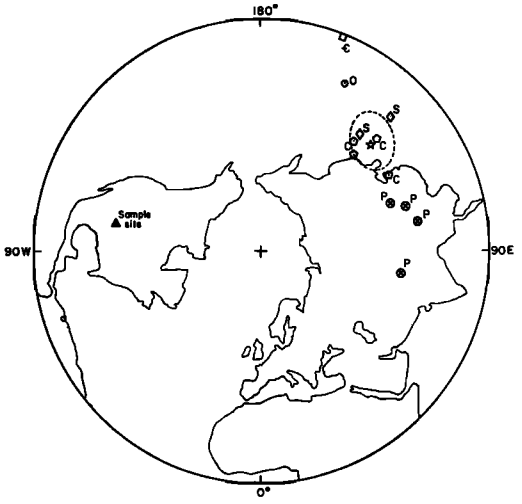


Fig. 5. Location of the mean virtual geomagnetic pole position of the Wichita Mountains basement rocks compared with Paleozoic VGP's for the United States [after *Irving*, 1964]. Mean VGP for Wichita Mountains (circle 33.5°N, 134°E) and the 95% confidence limit (dashed ellipse: $p = 7.6°$, $m = 14.9°$) with sample site (triangle) at 34.8°N, 99.9°W. Paleozoic VGP's (Є = Cambrian, O = Ordovician, S = Silurian, C = Carboniferous, and P = Permian).

was acquired not earlier than 520 to 560 m.y. ago. The final calculated VGP (reverse polarity) for the grouped samples, 33.5°N and 134°E, is plotted in Figure 5. It correlates best with Carboniferous VGP's for North American rocks [*Irving*, 1964]. It is therefore assumed that the low-temperature alteration of the iron oxides in the stable and grouped samples took place in Carboniferous time, that it can be correlated with the geological event of the uplift of the Wichita Mountains, and that it supports the interpretation by *Ham et al.* [1965] that the uplift took place in Late Pennsylvanian time.

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